Module 4

The Time Value of Money

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

Goals

- **Part 1**
  - Translate dollars today into dollars tomorrow (“future value”)
  - Translate dollars tomorrow into dollars today (“present value”)

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  - Financial calculators

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- **Part 3**
  - Special cases: perpetuities, annuities, and their variants

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- **Part 1**
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  - Translate dollars tomorrow into dollars today (“present value”)
- **Part 2**
  - Financial calculators
- **Part 3**
  - Special cases: perpetuities, annuities, and their variants
- **Part 4**
  - The role of inflation

Module 4, Part 1

Future Value and Present Value

The Time Value of Money



Would you rather receive a dollar today, or a dollar tomorrow?

The Time Value of Money

- A dollar today is worth more than a dollar tomorrow



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!
  - Assuming, there is some investment with a positive return for us to pursue



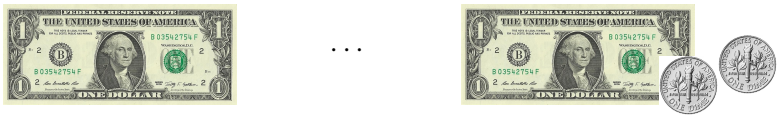
Future Value

- **Future value**, the amount to which an investment will grow after earning interest
- A "present value" today grows to its "future value" tomorrow via interest

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Today (time passes) Later



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$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

$$\text{Future value (FV)} = \$100 \times (1 + 0.06)^5 = \$133.82$$

Compound Interest

- **Compound interest**, the interest you earn with compounding, that is, with interest on interest

Future Value, Compound Interest

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- How much interest will you have earned after five years?

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Year	Start balance	Compound interest	End balance
1	\$100.00	$0.06 \times 100.00 = \$6.00$	\$106.00
2	\$106.00	$0.06 \times 106.00 = \$6.36$	\$112.36
3	\$112.36	$0.06 \times 112.36 = \$6.74$	\$119.10
4	\$119.10	$0.06 \times 119.10 = \$7.15$	\$126.25
5	\$126.25	$0.06 \times 126.25 = \$7.57$	\$133.82

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- With compound interest, your \$100 grows to $\$100(1 + r)^T = 133.82$, so you earned \$33.82 in compound interest

Compound Interest

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Compound Interest

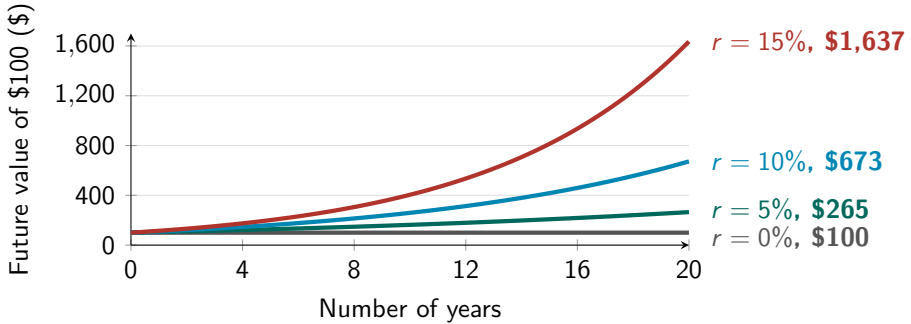
- **Compound interest**, the interest you earn with compounding, that is, with interest on interest
  - If you earn r per year for T years, your money grows to $\text{PV} \times (1 + r)^T$
  - The total compound interest you have earned is $\text{PV} \times (1 + r)^T - \text{PV}$

Compound Interest

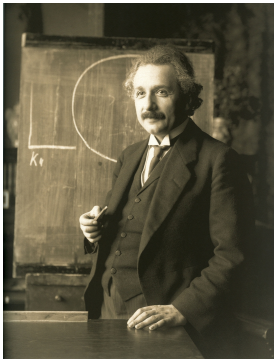
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  - The total compound interest you have earned is $\text{PV} \times (1+r)^T - \text{PV}$
- Compound interest is the default in finance (e.g., deposit rate)

Future Value Grows Fast in r and T

- The future value (FV) depends heavily on the interest rate r, especially as the time dimension T grows

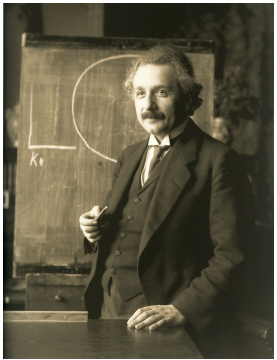


What's the Most Powerful Force in the Universe?



- Einstein, . . .

What's the Most Powerful Force in the Universe?



- Einstein, ...
- "Compound interest"

Example, Future Value

- Peter Minuit bought Manhattan Island for $24 in 1626
- a. What would that investment have been worth in 2021, if it had earned an 8% annual return?
- b. What if the return had been 3.5%?



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(roughly the same as total world wealth)

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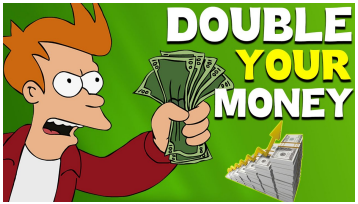
a. $FV = \$24 \times (1 + 0.08)^{395} = \$382$ trillion

(roughly the same as total world wealth)

b. $FV = \$24 \times (1 + 0.035)^{395} = \$19$ million

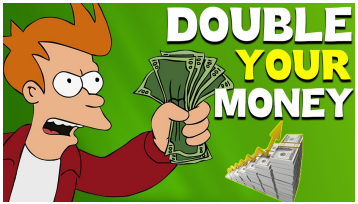
Example, Future Value

- Suppose you earn an annual return of 6%. How many years will it take until your money doubles?



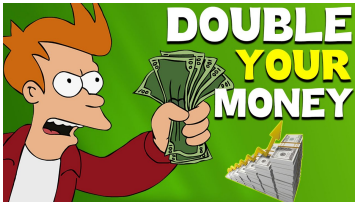
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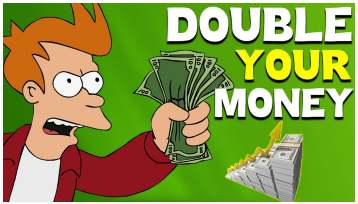
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  - We will learn how to calculate this with the financial calculator



Example, Future Value



$N = ?, \quad I = 6\%$

Example, Future Value



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- -1 **PV**, 6 **I/Y**, 0 **PMT**, 2 **FV**, then **CPT** **N**

Example, Future Value



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- The output is 11.9

Present Value

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- Recall, Future value $(\text{FV}) = \text{PV} \times (1 + r)^T$, that means

$$\text{Present value } (\text{PV}) = \text{FV} \times \frac{1}{(1 + r)^T}$$

Present Value

- **Present value**, the value today of a future cash flow
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$$\text{Present value (PV)} = \text{FV} \times \frac{1}{(1 + r)^T}$$

- \$100 in the future is worth less than \$100 in the present, so we "discount" the value of the future cash flow to its present value

Present Value

- **Discount rate**, the rate used to compute PVs (i.e., r)
  - We choose a discount rate that reflects the risk in the cash flow, which we will return to in **Modules 8 and 9**

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  - Less than one when r is positive

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Discount factor

- **Discounted cash flow (DCF)**, the method of calculating present values by discounting future cash flows

Example, Present Value

- Suppose you need $3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

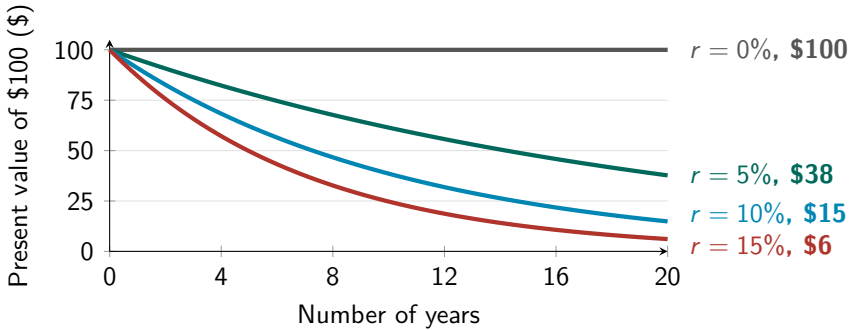
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$$\text{Present value (PV)} = \frac{\text{FV}}{(1+r)^T} = \frac{\$3{,}000}{(1+0.08)^1} = \$2{,}778$$

Present Value

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Present Value, Multiple Future Cash Flows

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$$\text{PV}_{\text{total}} = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \cdots$$

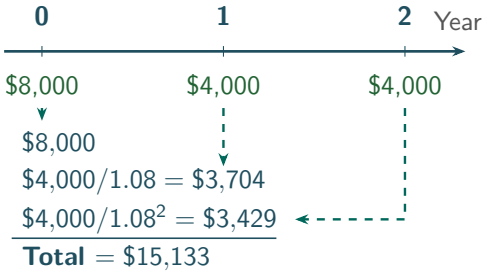
- where C_1 is the cash flow in one year, C_2 is the cash flow in two years, etc.

Example, Multiple Future Cash Flows

- You face two investments with different cash flows in the future
  - A. Receive $15,000 cash now, or
  - B. Receive three payments, $8,000 now, $4,000 one year from now, and $4,000 two years from now
- If the discount rate is 8%, which do you prefer?

Example, Multiple Future Cash Flows

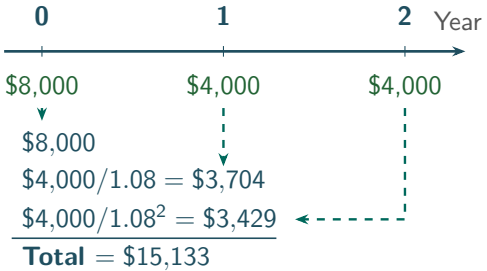
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Example, Multiple Future Cash Flows

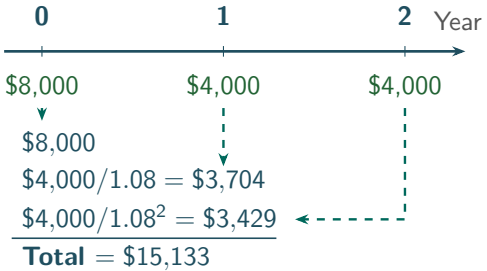
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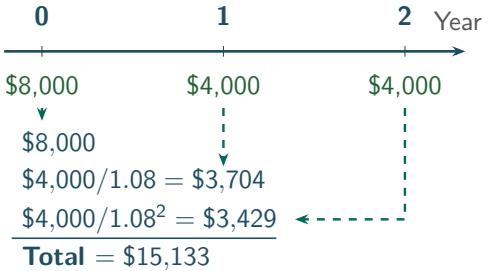


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- The present value of option B (\$15,133) is more than option A's \$15,000, so choose option B

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$$= \$15{,}133$$

- The present value of option B (\$15,133) is more than option A's \$15,000, so choose option B
- This is the principle behind the NPV rule in capital budgeting, which we will return to in **Modules 6 and 7**

Return

- **Return**, percentage growth of an asset's value over some period of time

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$$\text{Return} = \frac{\$180 - \$100}{\$100} = 80\%$$

Holding Period Return

- **Holding period return**, percentage growth over the period an asset is owned, which can be longer or shorter than a year, e.g., owning a stock for T periods with returns $r_1, r_2, \ldots, r_T$ gives

$$1 + r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) \times \cdots (1 + r_T)$$

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$$r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) - 1 = 1.80 \times 0.80 - 1 = 44\%$$

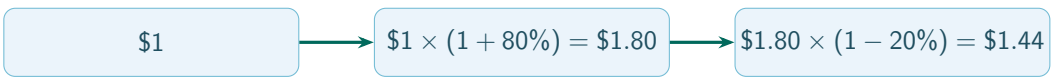
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Annualized Return

- **Annualized return**, percentage growth of money per year, calculated as a geometric mean over the T years of the holding period

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$$r_{\text{annualized}} = (1 + r_{\text{holding period}})^{1/T} - 1 = 1.44^{0.5} - 1 = 20\%$$

Annualized Return

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$$r_{\text{annualized}} = (1 + r_{\text{holding period}})^{1/T} - 1 = 1.44^{0.5} - 1 = 20\%$$

- The annualized return is *not* $\dfrac{80\% - 20\%}{2} = 30\%$

Recap, Three Kinds of Returns

Return	Holding period return	Annualized return
Percentage growth of an asset's value over some period of time	Percentage growth over the whole period an asset is owned, longer or shorter than a year	Percentage growth per year, a geometric mean over the T years of the holding period

Recap, Three Kinds of Returns

Return	**Holding period return**	**Annualized return**
Percentage growth of an asset's value over some period of time	Percentage growth over the whole period an asset is owned, longer or shorter than a year	Percentage growth per year, a geometric mean over the T years of the holding period

- These financial returns are *not directly related in any way* with **Module 3**'s accounting-based ROA and ROE measures

Recap of Part 1

- A dollar today is worth more than a dollar tomorrow

Recap of Part 1

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- **Future value**, the amount to which an investment will grow after earning interest

$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

Recap of Part 1

- A dollar today is worth more than a dollar tomorrow
- **Future value**, the amount to which an investment will grow after earning interest

$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

- **Present value**, the value of a future cash flow multiplied by its discount factor

$$\text{Present value (PV)} = \text{FV} \times \frac{1}{(1 + r)^T}$$

Recap of Part 1

- A dollar today is worth more than a dollar tomorrow
- **Future value**, the amount to which an investment will grow after earning interest

$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

- **Present value**, the value of a future cash flow multiplied by its discount factor

$$\text{Present value (PV)} = \text{FV} \times \frac{1}{(1 + r)^T}$$

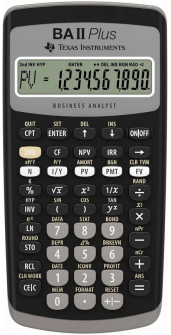
- Present values can be added together to calculate the total present value of multiple future cash flows

Module 4, Part 2

Financial Calculators

Get the Texas Instruments BA II Plus

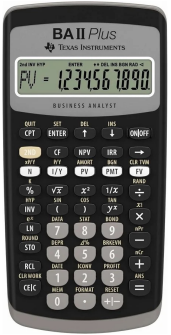
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CFA

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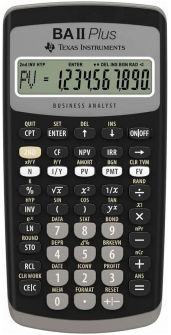
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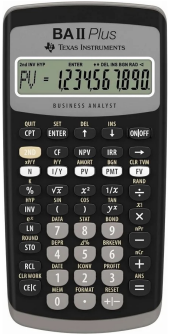
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- You need a **financial calculator**
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- CFA (Chartered Financial Analyst), a key certification, often required in finance jobs
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- After you reset, you should repeat the next two settings!

Set the Number of Digits to at Least 4!

- On the TI BA II Plus, press
  - **2nd**
  - **.**
  - 4
  - **ENTER**

Set the Number of Compounding Periods per Year to 1!

- On the TI BA II Plus, press
  - **2nd**
  - **I/Y** (**P/Y**)
  - 1
  - **ENTER**

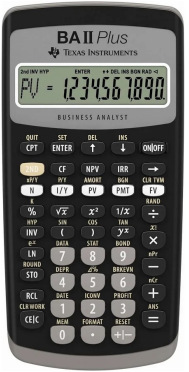
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You must do this!

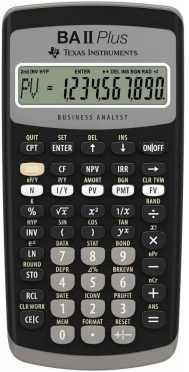
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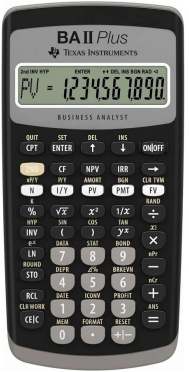
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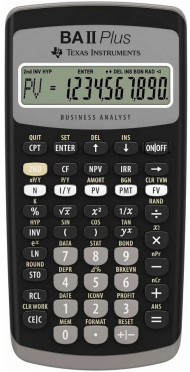
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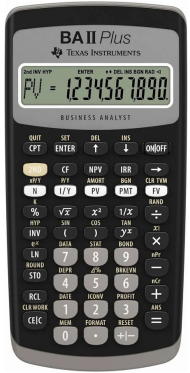
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- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

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- Why does the answer come out negative?

The Equation the Calculator Solves

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- In 2007, Puerto Rico sold bonds that promised to pay $1,000 after 47 years (no periodic "coupon" payments)
- These bonds were sold at $94.40 each
- What was the interest rate paid by Puerto Rico?



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$N = 47, \quad I = ?$

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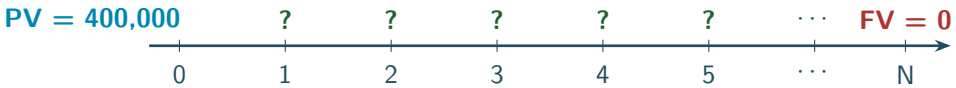
The present values of PV, all PMTs, and FV, summed up together equal 0.

Example, Mortgage

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Example, Mortgage



$N = 30, \quad I = 4.8\%$

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- That is, you would pay an undiscounted total of $30 \times 25{,}430$ over the course of your mortgage

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- Financial calculators solve any one of **N**, **I/Y**, **PV**, **PMT**, and **FV** from the other four

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- Set the number of compounding periods per year to 1, and show at least 4 digits

Module 4, Part 3

Special Cases: Perpetuities and Annuities

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- Why care about perpetuities and annuities?
  - Many real-world cash flows have these patterns (particularly annuities, e.g., mortgages)
  - Neat shortcuts make it very easy to calculate the present value of these cash flows

Perpetuity Structure

- A perpetuity is a stream of equal cash flows (C) that occur at regular intervals and last forever



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- A perpetuity has a present value of

$$\text{PV of perpetuity} = \frac{\text{Cash flow}}{\text{Discount rate}} = \frac{C}{r}$$

Example, Perpetuities

- Suppose the San Antonio Spurs offer Victor Wembanyama a perpetual payment of $20m every year. He will be paid forever, as long as he lives, and then his children, and then his children's children, etc.
- Suppose the discount rate is 10%. What is the present value of this contract?



Example, Perpetuities

- Solution,

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$$\text{PV} = \frac{C}{r} = \frac{20}{10\%} = \$200m$$

Perpetuity, Timing

- **Perpetuity due**, first payment at time 0

$$\text{PV} = \frac{C}{r} \times (1 + r)$$



Perpetuity, Timing

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- **Ordinary perpetuity**, first payment at time 1

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- **Delayed perpetuity**, first payment at time $t > 1$

$$\text{PV} = \frac{C}{r} \times \frac{1}{(1+r)^{t-1}}$$

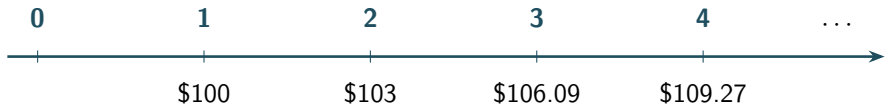


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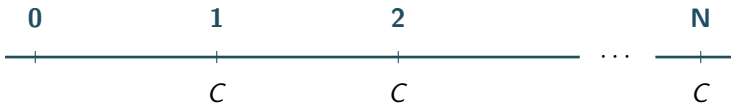


$$\text{PV of growing perpetuity} = \frac{\text{Cash flow}}{\text{Discount rate} - \text{Growth rate}} = \frac{C}{r - g}$$

- This formula comes up often when valuing stocks, e.g., in the "Gordon growth model" and DCF valuation (more in **Module 5**)
- When $g = 0$, the PV simplifies to the formula for a regular (non-growing) perpetuity

Annuities

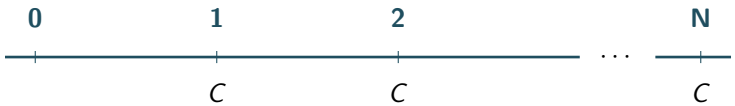
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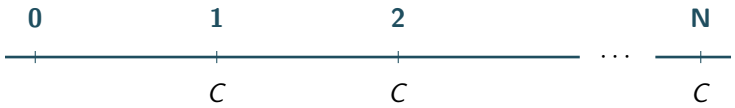


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- Present value of an annuity,

$$\text{PV of annuity} = C \underbrace{\frac{1}{r}\left(1 - \frac{1}{(1+r)^N}\right)}_{\substack{\text{present value annuity factor} \\ \text{(PVAF)}}}$$

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- C is the annual cash flow (aka the payment), r is the discount rate, and N is the number of years that the annuity lasts

Valuing Annuities with a Financial Calculator

- The financial calculator keys map onto an annuity,
  - **N**, the number of payments
  - **I/Y**, the discount rate
  - **PV**, the present value of the annuity
  - **PMT**, the recurring payment (the C in the formula)
  - **FV**, 0 when there is no single payment at the end

Example, Annuity

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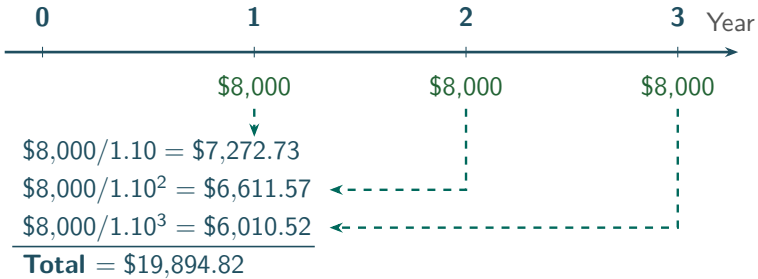
- On a financial calculator, **PMT** $= -8{,}000$, **N** $= 3$, **I/Y** $= 10$, **FV** $= 0$, leading to \$19,895

Example, Annuity

- We can verify this by discounting each cash flow separately and adding them up

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Discuss, Front-Loaded or Constant Cash Flows?

- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?

Example, Front-Loaded or Constant Cash Flows

- Two streams, the same $24,000 undiscounted total, discount rate 10%

Cash flows	Year 1	Year 2	Year 3	Total	PV at 10%
Annuity (the car)	$8,000	$8,000	$8,000	$24,000	?
Front-loaded	$12,000	$8,000	$4,000	$24,000	?

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- Annuity, $\$8{,}000/1.10 + \$8{,}000/1.10^2 + \$8{,}000/1.10^3 = \$19{,}895$
- Front-loaded, $\$12{,}000/1.10 + \$8{,}000/1.10^2 + \$4{,}000/1.10^3 = \$20{,}526$

Discuss, Front-Loaded or Constant Cash Flows?

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- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?
- Answer, the front-loaded one
- What about a back-loaded cash flow and an annuity?

Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

$$\text{PV} = \text{PV annuity} \times (1 + r)$$



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- **Ordinary annuity**, first payment at time 1 $\text{PV} = \text{PV annuity}$



- **Delayed annuity**, first payment at time $t > 1$ $\text{PV} = \text{PV annuity} \times \frac{1}{(1+r)^{t-1}}$



Practice Problem

- A store sells a product for \$100. It offers you two payment plans,
  - Plan A, you pay \$25 today and \$25 at the end of each of the next 3 years
  - Plan B, you pay the entire bill today and get a 7% discount, so you pay only \$93

a. Suppose the interest rate is 4%. What is the present value of your payments under Plan A?

b. Which plan is a better deal?

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a. Suppose the interest rate is 4%. What is the present value of your payments under Plan A?

b. Which plan is a better deal?

We choose the payment plan with the smaller PV.

Practice Problem

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Practice Problem

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- Plan B, pay the entire bill today, \$93 after the 7% discount



Practice Problem

- a. This is an annuity due, since the first payment is due today

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- Value the 4 payments as an ordinary annuity (first payment at time 1),

$$\text{PV annuity} = \$25 \times \frac{1}{0.04}\left(1 - \frac{1}{(1+0.04)^4}\right) = \$90.75$$

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- b. Paying in full now is better ($93 beats $94.38)
- On a financial calculator, **PMT** $= -25$, **N** $= 4$, **FV** $= 0$, **I/Y** $= 4$, leading to 90.75, and next, $90.75 \times 1.04 = 94.38$

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- Timing,
  - Ordinary perpetuities and annuities make their first payment at time $t = 1$
  - Perpetuity and annuity dues make their first payment at $t = 0$
  - Delayed perpetuities and annuities make their first payment at $t > 1$

Module 4, Part 4

Inflation

Motivating Example, Pandemic-Era Inflation

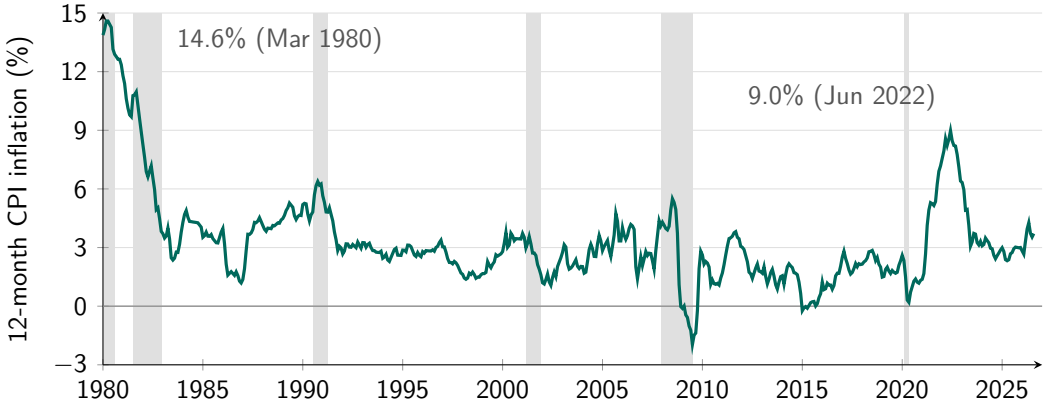


WSJ, July 13, 2022 (5:01), "How the Consumer-Price Index Measures Inflation"

Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)
  - **Consumer price index (CPI)**, the price of a "basket" of goods that a typical household buys

History of Inflation



Source, BLS via FRED (CPI-U, all items), shaded areas are NBER recessions

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- **Real return**, the rate at which purchasing power increases
  - Compare the ability to buy a basket of goods at time $t = 0$ to the ability to buy the same basket of goods at time $t = 1$

Inflation and Returns

- In the presence of inflation, an investor's real return is always less than the nominal return

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$$\text{Approximation,} \quad \text{Real return} \approx \text{Nominal return} - \text{Inflation rate}$$

Example, Buying Apples

- Suppose we live on a planet where people only eat apples

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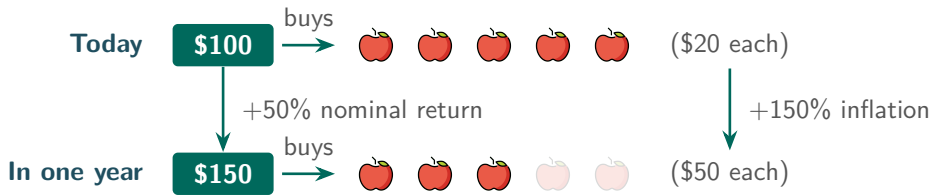
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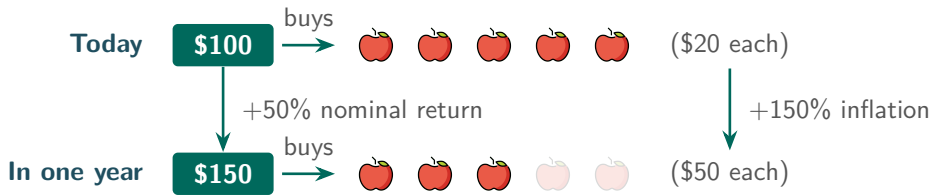
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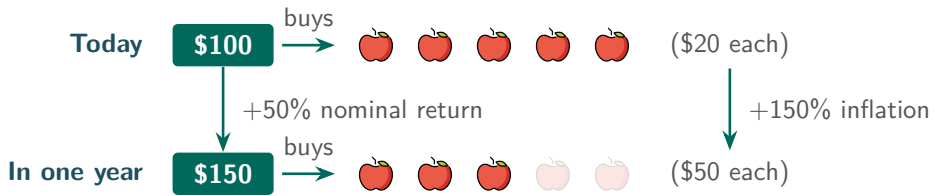
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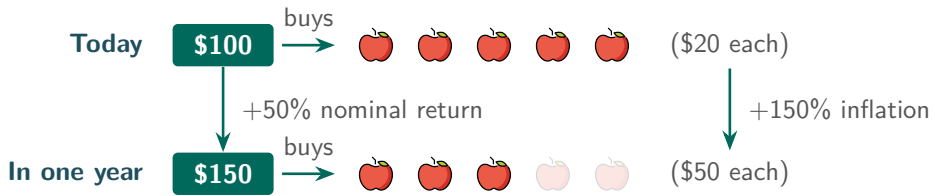


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- The number of apples you can buy drops by 40% relative to before

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- Approximation, real $\approx 6\% - 2\% = 4\%$

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- $(1 + \text{Real return}) = (1 + \text{Nominal return})/(1 + \text{Inflation rate})$, and approximately, real return $\approx$ nominal return $-$ inflation rate

Practice Problem

- You have \$20,000 in the bank earning interest of 5% per year. You need \$30,000 to make a down payment on a house. You can save an additional \$1,000 per year, deposited at the end of each year. How long will it take you to accumulate the \$30,000?

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$I = 5\%, \quad N = ?$

Practice Problem

- Solution, we need to solve for **N**,
  - $-20{,}000$ **PV** (the money already in the bank)
  - $-1{,}000$ **PMT** (the yearly deposits)
  - $30{,}000$ **FV** (the target, with the opposite sign)
  - 5 **I/Y**
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Practice Problem

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- The calculator is solving

$$-\$20{,}000 - \left[\frac{\$1{,}000}{(1+0.05)^1} + \cdots + \frac{\$1{,}000}{(1+0.05)^N}\right] + \frac{\$30{,}000}{(1+0.05)^N} = 0$$

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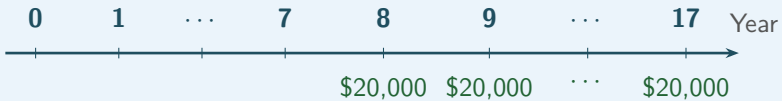
- The output is 4.57, so it takes about 4.6 years to accumulate the \$30,000

Practice Problem

- You will receive annual payments of $20,000 for a period of 10 years. The first payment will be made 8 years from now. The interest rate is 6% per year. How much is the present value of this stream of payments?

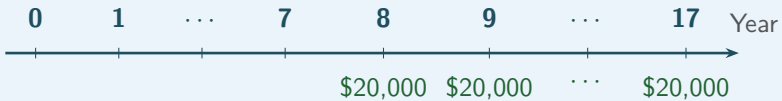
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10 payments in all, the first at year 8 and the last at year 17

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- On a financial calculator, two steps,
  - Step 1, 10 **N**, 6 **I/Y**, −20,000 **PMT**, 0 **FV**, then **CPT** **PV**, the output is 147,202 (the value at year 7)

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  - Step 2, 7 **N**, 6 **I/Y**, 0 **PMT**, −147,202 **FV**, then **CPT** **PV**, the output is \$97,898

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- The real return strips inflation out of the nominal return, real return $\approx$ nominal return $-$ inflation rate

Thank you!

Please let me know if you have any questions.